

EX-6. CALCULATE FREQUENCY OF GIVEN WORD IN A FILE USING MAPREDUCE

STEP:1 Starting Namenode and Datanode

```
C:\Hadoop1\hadoop-3.5.0\sbin>start-dfs.cmd
```

```
C:\Hadoop1\hadoop-3.5.0\sbin>jps
```

16544 DataNode

16460 Jps

8348 NameNode

STEP:1 Starting ResourceManager and NodeManager

```
C:\Hadoop1\hadoop-3.5.0\sbin>start-yarn.cmd
```

starting yarn daemons

```
C:\Hadoop1\hadoop-3.5.0\sbin>jps
```

16544 DataNode

13892 Jps

11800 ResourceManager

19656 NodeManager

8348 NameNode

STEP:2

1. Open Notepad.
2. Enter some text, for example:

Hadoop is a big data framework.
Hadoop is used for big data processing.
Hadoop provides distributed storage.

3. Click File → Save As.
4. Save the file as:

```
C:\Hadoop1\sample.txt
```

Make sure Save the filename is **sample.txt**.

5. Check that the file exists:

```
dir C:\Hadoop1\sample.txt
```

STEP:3 First create the HDFS input directory:

C:\Hadoop1\hadoop-3.5.0\sbin>hdfs dfs -mkdir -p /input

STEP:4 Then upload your file:

C:\Hadoop1\hadoop-3.5.0\sbin>hdfs dfs -put C:\Hadoop1\sample.txt /input

STEP:5 Verify the file

hdfs dfs -ls /input

STEP:6 To see the contents of file created

hdfs dfs -cat /input/sample.txt

STEP:7: Open Notepad

- create a new file, and save as WordFrequency.java in location (C:\Hadoop1\WordFrequency.java)
- Paste this Java program

```
import java.io.IOException;
import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.Mapper;
import org.apache.hadoop.mapreduce.Reducer;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;
```

```
public class WordFrequency {
    public static class TokenizerMapper
        extends Mapper<Object, Text, Text, IntWritable> {

        private final static IntWritable one = new IntWritable(1);
        private Text word = new Text();

        public void map(Object key, Text value, Context context)
            throws IOException, InterruptedException {

            String[] words = value.toString().split("\\s+");

            for (String str : words) {
```

```

        word.set(str);
        context.write(word, one);
    }
}

}

public static class IntSumReducer
    extends Reducer<Text, IntWritable, Text, IntWritable> {

    public void reduce(Text key, Iterable<IntWritable> values,
        Context context)
        throws IOException, InterruptedException {

        int sum = 0;

        for (IntWritable value : values) {
            sum += value.get();
        }

        context.write(key, new IntWritable(sum));
    }
}

public static void main(String[] args) throws Exception {
    Configuration conf = new Configuration();

    Job job = Job.getInstance(conf, "Word Frequency");

    job.setJarByClass(WordFrequency.class);
    job.setMapperClass(TokenzierMapper.class);
    job.setReducerClass(IntSumReducer.class);

    job.setOutputKeyClass(Text.class);
    job.setOutputValueClass(IntWritable.class);

    FileInputFormat.addInputPath(job, new Path(args[0]));
    FileOutputFormat.setOutputPath(job, new Path(args[1]));

    System.exit(job.waitForCompletion(true) ? 0 : 1);
}
}

```

STEP:8 Check that the file exists
dir C:\Hadoop1\WordFrequency.java

STEP:8 Compile the java file

```
javac -classpath  
"%HADOOP_HOME%\share\hadoop\common\*; %HADOOP_HOME%\share\hadoop\comm  
on\lib\*" -d . WordFrequency.java
```

///IF ERROR

STEP:9 First check your Hadoop MapReduce JARs

C:\Hadoop1>

run:

dir C:\Hadoop1\hadoop-3.5.0\share\hadoop\mapreduce

You should see files such as:

```
hadoop-mapreduce-client-core-3.5.0.jar  
hadoop-mapreduce-client-common-3.5.0.jar  
Hadoop-mapreduce-client-jobclient-3.5.0.jar
```

RUN:

```
javac -classpath "C:\Hadoop1\hadoop-  
3.5.0\share\hadoop\common\*;C:\Hadoop1\hadoop-  
3.5.0\share\hadoop\common\lib\*;C:\Hadoop1\hadoop-  
3.5.0\share\hadoop\hdfs\*;C:\Hadoop1\hadoop-  
3.5.0\share\hadoop\mapreduce\*;C:\Hadoop1\hadoop-3.5.0\share\hadoop\yarn\*" -d .  
WordFrequency.java
```

C:\Hadoop1>WordFrequency.class

C:\Hadoop1>WordFrequency\$TokenizerMapper.class

C:\Hadoop1>WordFrequency\$IntSumReducer.class

C:\Hadoop1>jar -cvf WordFrequency.jar *.class

C:\Hadoop1>hdfs dfs -ls /input

Found 1 items

-rw-r--r-- 1 VIJAY supergroup 110 2026-08-24 11:45 /input/sample.txt

```
C:\Hadoop1>hdfs dfs -cat /input/sample.txt
```

Hadoop is a big data framework.

Hadoop is used for big data processing.

Hadoop provides distributed storage.

```
C:\Hadoop1>hadoop jar WordFrequency.jar WordFrequency /input /output
```

OUTPUT:

```
C:\Hadoop1>hdfs dfs -cat /output/part-r-00000
```

Hadoop 3

a 1

big 2

data 2

distributed 1

for 1

framework. 1

is 2

processing. 1

provides 1

storage. 1

used 1